HIGHER PRO-ETALE COHOMOLOGY OF THE PULLBACK OF AN INJECTIVE SHEAF ON AN AFFINE SCHEME VANISHES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the verification of the first two conditions of the Cech criterion, the reduction of a pro-etale covering to a filtered colimit of etale coverings, the approximation of an etale covering along an inverse limit of etale neighbourhoods, and the closing acyclicity argument.

Editorial note. The label lemma-affine-vanishing names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-affine-vanishing –start 4082 –end 4167 –local-id 3576 returned RESULT=OK (L3576 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json (new exclusive pool assigned by the Lead on 2026-10-13).

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The Cech vanishing criterion for a collection of objects and coverings. Cohomology on Sites, the lemma giving conditions (1), (2), (3) on a set of objects and a set of coverings under which the higher Cech cohomology of an injective sheaf vanishes. Supplies the criterion applied to the set of affine objects whose structure rings are ind-etale over the ring of X and to the pro-etale coverings between them.
- Permanence and base change for ind-etale algebras, and affineness of the pro-etale site. Same chapter, the lemmas on composition and base change of ind-etale algebras and the lemma that an affine scheme is a final object with the expected coverings in the pro-etale topology, together with the proposition on weakly etale morphisms. Supply conditions (1) and (2) of the criterion.
- Comparison of the pullback along the pro-etale to etale morphism with restriction. Same chapter, the lemma identifying the pullback of an etale sheaf to the pro-etale site with its restriction, and the lemma computing finite limits of the resulting sections. Supply the identification of the Cech complexes of the pullback with those of the original sheaf over the approximating etale coverings.

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- Acyclicity of the Cech complex of an injective sheaf. Cohomology on Sites, the lemma that the Cech complex of an injective sheaf over a covering is acyclic in positive degrees. Supplies the vanishing of the Cech cohomology of the approximating etale coverings, which passes to the limit.
- Descent of finite presentation, affine schemes and etale morphisms along a directed limit. Limits, the lemmas descending finite presentation, affine schemes and etale morphisms along a directed inverse limit. Supply the approximation of the etale covering of U by an etale covering of one U_b.

Source reference. Original reference 097J: sources/stacks-proetale/pinned-original-source.tex, source lines 1387–1387. Prerequisite pointer invoked at proof line(s) 4101; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 097K: sources/stacks-proetale/pinned-original-source.tex, source lines 1398–1398. Prerequisite pointer invoked at proof line(s) 4100; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0241: sources/stacks-proetale/pinned-original-source.tex, source lines 4045–4045. Prerequisite pointer invoked at proof line(s) 4159; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 099S: sources/stacks-proetale/pinned-original-source.tex, source lines 4058–4058. Prerequisite pointer invoked at proof line(s) 4129; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 098D: sources/stacks-proetale/pinned-original-source.tex, source lines 2433–2433. Prerequisite pointer invoked at proof line(s) 4102; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07RP: sources/stacks-chow/tags.txt, source lines 9874–9874. Prerequisite pointer invoked at proof line(s) 4144; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01ZM: sources/stacks-chow/tags.txt, source lines 2599–2599. Prerequisite pointer invoked at proof line(s) 4142; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 07RR: sources/stacks-chow/tags.txt, source lines 9876–9876. Prerequisite pointer invoked at proof line(s) 4152; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 01Z6: sources/stacks-chow/tags.txt, source lines 2581–2581. Prerequisite pointer invoked at proof line(s) 4143; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 097Z: sources/stacks-proetale/pinned-original-source.tex, source lines 1802–1802. Prerequisite pointer invoked at proof line(s) 4103; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03F9: sources/stacks-chow/tags.txt, source lines 4484–4484. Prerequisite pointer invoked at proof line(s) 4091; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03AW: sources/stacks-chow/tags.txt, source lines 4325–4325. Prerequisite pointer invoked at proof line(s) 4163; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be an affine scheme. For injective abelian sheaf $\mathcal{I}$ on $X_{étale}$ we have $H^p(X_{pro\text{-}étale}, \epsilon^{-1}\mathcal{I}) = 0$ for $p > 0$.*

Proof. We are going to use Cohomology on Sites, Lemma 03F9 to prove this. Let $\mathcal{B} \subset \mathrm{Ob}(X_{pro\text{-}étale})$ be the set of affine objects U of $X_{pro\text{-}étale}$ such that $\mathcal{O}(X) \to \mathcal{O}(U)$ is ind-étale. Let Cov be the set of pro-étale coverings $\{U_i \to U\}_{i=1,\ldots,n}$ with $U \in \mathcal{B}$ such that $\mathcal{O}(U) \to \mathcal{O}(U_i)$ is ind-étale for $i = 1, \ldots, n$. Properties (1) and (2) of Cohomology on Sites, Lemma 03F9 hold for $\mathcal{B}$ and Cov by Lemmas 097K, 097J, and 098D and Proposition 097Z.

To check condition (3) suppose that $\mathcal{U} = \{U_i \to U\}_{i=1,\ldots,n}$ is an element of Cov. We have to show that the higher Cech cohomology groups of $\epsilon^{-1}\mathcal{I}$ with respect to $\mathcal{U}$ are zero. First we write $U_i = \lim_{a \in A_i} U_{i,a}$ as a directed inverse limit with $U_{i,a} \to U$ étale and $U_{i,a}$ affine. We think of $A_1 \times \ldots \times A_n$ as a direct set with ordering $(a_1, \ldots, a_n) \geq (a'_1, \ldots, a'_n)$ if and only if $a_i \geq a'_i$ for $i = 1, \ldots, n$. Observe that $\mathcal{U}_{(a_1,\ldots,a_n)} = \{U_{i,a_i} \to U\}_{i=1,\ldots,n}$ is an étale covering for all $a_1, \ldots, a_n \in A_1 \times \ldots \times A_n$. Observe that

$$U_{i_0} \times_U U_{i_1} \times_U \ldots \times_U U_{i_p} = \lim_{(a_1,\ldots,a_n) \in A_1 \times \ldots \times A_n} U_{i_0,a_{i_0}} \times_U U_{i_1,a_{i_1}} \times_U \ldots \times_U U_{i_p,a_{i_p}}$$

for all $i_0, \ldots, i_p \in \{1, \ldots, n\}$ because limits commute with fibred products. Hence by Lemma 099S and exactness of filtered colimits we have

$$\check{H}^p(\mathcal{U}, \epsilon^{-1}\mathcal{I}) = \mathrm{colim}\, \check{H}^p(\mathcal{U}_{(a_1,\ldots,a_n)}, \epsilon^{-1}\mathcal{I})$$

Thus it suffices to prove the vanishing for étale coverings of U!

Let $\mathcal{U} = \{U_i \to U\}_{i=1,\ldots,n}$ be an étale covering with U_i affine. Write $U = \lim_{b \in B} U_b$ as a directed inverse limit with U_b affine and $U_b \to X$ étale. By Limits, Lemmas 01ZM, 01Z6, and 07RP we can choose a $b_0 \in B$ such that for $i = 1, \ldots, n$ there is an étale morphism $U_{i,b_0} \to U_{b_0}$ of affines such that $U_i = U \times_{U_{b_0}} U_{i,b_0}$. Set $U_{i,b} = U_b \times_{U_{b_0}} U_{i,b_0}$ for $b \geq b_0$. For b large enough the family $\mathcal{U}_b = \{U_{i,b} \to U_b\}_{i=1,\ldots,n}$ is an étale covering, see Limits, Lemma 07RR. Exactly as before we find that

$$\check{H}^p(\mathcal{U}, \epsilon^{-1}\mathcal{I}) = \mathrm{colim}\, \check{H}^p(\mathcal{U}_b, \epsilon^{-1}\mathcal{I}) = \mathrm{colim}\, \check{H}^p(\mathcal{U}_b, \mathcal{I})$$

the final equality by Lemma 0241. Since each of the Čech complexes on the right hand side is acyclic in positive degrees (Cohomology on Sites, Lemma 03AW) it follows that the one on the left is too. This proves condition (3) of Cohomology on Sites, Lemma 03F9. Since $X \in \mathcal{B}$ the lemma follows. □